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FEDERAL STANDARD

FILM, PHOTOGRAPHIC, BLACK AND WHITE, CLASSIFICATION AND TESTING METHODS

This standard was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal Agencies.

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INFORMATION SHEET

ON

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New and revised pages will be issued under Change Notices which will be numbered consecutively and will bear the date of issuance. Change Notices should be retained and filed in front of the Standard until such time as they are superseded by a reissue of the entire Standard.

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FEDERAL STANDARD

FILM, PHOTOGRAPHIC, BLACK AND WHITE, CLASSIFICATION AND TESTING METHODS

AUTHORITY. This standard is issued pursuant to the Federal Property and Administrative Services Act of 1949, as amended, and its application to the purchase of commodities referred to herein is mandatory on all Federal agencies.

S1. SCOPE

S1.1 Scope. This standard provides uniform testing methods and standards for determining the characteristics of sensitized photographic films, of the gelatin-silver halide-type emulsions, coated on flexible transparent plastic supports, and having as an end product black-and-white positive or negative photographic images.

S1.1.1 Classification. This standard also defines the classification system set up according to the normal use of the product, support characteristics, spectral sensitivity, gamma, and relative sensitivity when tested in accordance with this standard and the detail specification.

S1.2 Purpose. The purpose of this standard is to:

a. Provide classification of photographic films.
b. Provide definition of test terms.
c. Establish methods of testing.
d. Prescribe and describe test equipment.
e. Provide standards of measurement for the characteristics of the film.

S2. REFERENCED DOCUMENTS

S2.1 Specifications and standards. The following specifications and standards, of the issues in effect on the date of invitation for bids or request for proposal, form a part of this standard to the extent specified herein.

FEDERAL

Specifications

O-C-275 – Chemicals, Photographic (Single Substance) General Specification For

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Standards

FED. TEST METHOD
STD. NO. 406 — Plastics: Methods of Testing

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

MILITARY

Specifications

MIL-D-4824 — Developer, Photographic, High Contrast, Armed Forces Developer No 1.
MIL-D-4826 — Developer, Photographic, Low Contrast, Armed Forces Developer No 3.
MIL-F-26615 — Fixing Bath, Conventional Photographic, Acid Hardening.

Handbooks

MIL-HDBK-25 — Glossary of Photographic Terms (Including Document Reproduction).

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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S2.2 Other publications. The following documents form a part of this standard to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

American National Standards Institute, Inc Standards (ANSI).

PH1.29	—	Curl of Photographic Film, Methods for Determining the.
PH1.31	—	Brittleness of Photographic Film, Methods for Determining the.
PH1.32	—	Determining the Dimensional Change Characteristics of Photographic Films and Papers, Methods for.
PH2.5	—	Speed of Photographic Negative Materials, (Monochrome, Continuous-Tone), Method for Determining the.
PH2.19	—	Diffuse Transmission Density.
PH2.29	—	Simulated Daylight Source for Photographic Sensitometry.
PH2.33	—	Resolving Power of Photographic Materials, Method for Determining the.
PH2.34	—	Speeds of Monochrome Photographic Negative Films for Aerial Photography, Method for Determining the.

(Application for copies should be addressed to the American National Standards Institute, Inc 1430 Broadway, New York N.Y. 10018.)

S3. DEFINITIONS

S3.1 Terms. For the purpose of this standard, all photographic terms not defined herein shall be in accordance with MIL-HDBK-25.

S3.1.1 Sensitometric terms.

S3.1.1.1 Density. Density shall be diffuse visual transmission density as defined in ANSI PH2.19.

S3.1.1.1.1 Total density. Total density shall be defined as the sum of the emulsion density and the base density as defined herein.

S3.1.1.1.2 Base density. Base density shall be defined as the density of a fixed-out strip prepared as specified herein.

S3.1.1.1.3 Net density. Net density shall be defined as the difference between total density and base density.

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S3.1.1.2 Fog. Fog shall be defined as the average net density of film processed as a negative that has not been exposed to light prior to development.

S3.1.1.3 Gamma. Gamma shall be as defined in MIL-HDBK-25. The "straight line portion" is selected where it most nearly approximates a straight line nearest to the toe of the curve.

S3.1.1.4 Average gradient. Average gradient shall be defined as the slope of the straight line drawn between specified points on the characteristic curve.

S3.1.1.5 Filter factor. The filter factor of a film shall be defined as the number of times exposure must be increased to compensate for light absorbed by a filter.

S3.1.1.6 Relative sensitivity. For purposes of this standard and applicable detail specifications "relative sensitivity" shall be that value determined by methods "A" through "F" or as stated in the detail specification.

S3.1.1.6.1 Method A. Method A relative sensitivity (negative) shall be as defined in ANSI PH2.5 (arithmetic method).

S3.1.1.6.2 Method B. Method B relative sensitivity (positive and reversal) shall be defined as 10/E, where E is the exposure (in lux-seconds) at the point on the characteristic curve where the density is 1.00 greater than the base plus fog density (or in the case of reversal films, base plus minimum density).

S3.1.1.6.3 Method C. Method C relative sensitivity (photomechanical and process) shall be defined as 10/E, where E is the exposure (in lux-seconds) at the point on the characteristic curve where the density is 2.50 greater than the base plus fog density.

S3.1.1.6.4 Method D. Method D relative sensitivity (microfilm) shall be defined as 45/E, where E is the exposure (in lux-seconds) at the point on the characteristic curve where the density is 1.20 greater than the base plus fog density.

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S3.1.1.6.5 Method E. Method E relative sensitivity (aerial film) shall be as defined in ANSI PH2.34.

S3.1.1.6.6 Method F. Method F relative sensitivity (infrared) shall be defined as 1/E, where E is the exposure (in ergs per square centimeter) at the point on the characteristic curve where the density is 0.20 greater than the base plus fog density.

S3.1.1.7 Relative sensitivity designator. The relative sensitivity designator shall be the alphabetical character used to identify the average relative sensitivity as determined by the above methods or as stated in the detail specification. Each alphabetical designator represents 0.3 log E increments in relative sensitivity. For the convenience of the procurement agencies and users, subdivisions of a designator may be necessary. These subdivisions shall be in 0.1 log E increments and shall be designated by the digits 1, 2, and 3 in increasing order of sensitivity; e.g., for designator "j" (limits 281 to 570) the subdivisions and limits would be as follows: j1 - 281 to 355, j2 - 356 to 450, j3 - 451 to 570.

S31.1.8 Emulsion number. An emulsion number shall be defined as a number assigned by the manufacturer to a uniform production run.

S3.1.2 Physical terms

S3.1.2.1 Physical defects. Physical defects shall be defined as scratches, pinholes, or other physical defects including streaks, spots, or areas of uneven coating.

S3.1.2.2 Film direction. The length direction of the film shall be defined as the direction parallel to its forward movement in the base casting machine. This is also termed "grain" or "machine direction". The width direction shall be at right angles to the machine direction in the plane of the sheet and is also termed "cross direction".

S3.1.2.3 Humidity coefficient of expansion. The humidity coefficient of expansion of a film shall be defined as the average change in dimension per unit distance per 1 percent change in relative humidity.

S3.1.2.4 Thermal coefficient of expansion. The thermal coefficient of expansion shall be defined as the average change in dimension per unit distance per degree Celsius (C) or Fahrenheit (F) change in temperature.

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S3.1.2.5 Dimensional change due to processing. Dimensional change due to processing shall be defined as the permanent dimensional change caused by wetting and drying of the sensitized material during photographic processing. It is measured after the processed and dried film has reached equilibrium with an atmosphere of the same relative humidity and temperature, as used in the premeasurement conditioning of the unprocessed film, and is expressed as a percentage.

S3.1.2.6 Dimensional change due to processing plus aging. Dimensional change due to processing plus aging shall be defined as the permanent dimensional change which occurs as a result of processing plus aging of the processed film. It is measured after the processed, aged film has reached equilibrium with an atmosphere of the same relative humidity and temperature, as used in the premeasurement conditioning of the unprocessed film, and is expressed as a percentage.

S3.1.2.7 Differential dimensional change. The differential dimensional change shall be the difference between the dimensional changes of the material in the two principal directions (machine and cross).

S3.1.2.8 Brittleness. Brittleness shall be defined as that property of the film which causes it to crack, break, or fail when deformed by bending.

S3.1.2.9 Tensile strength. Tensile strength shall be defined as the force, applied parallel to the plane of the film in tension, required to rupture a sample of film of specified width, length, and thickness under specified conditions of loading.

S3.1.2.10 Total thickness. Total thickness shall be defined as the thickness of the end product, including the emulsion, emulsion support, plus any undercoating, backing layers, or overcoatings as appropriate.

S3.1.2.11 Curl. Curl shall be defined as the deviation from flatness of the film plane. Curl may be caused by either dimensional differences between the support and other layers or by flow when the film is stored in a wound configuration.

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S3.1.2.12 Resolving power of photographic materials. The resolving power of photographic materials shall be as defined in ANSI PH2.33.

S4. FILM CLASSIFICATION SYSTEM

S4.1 Classification. All photographic film shall be classified in accordance with Table I. Limits are specified in Table I for the designators of total thickness, gamma, and relative sensitivity. These limits are used in conjunction with the respective values of a product for establishing the designator. The following example illustrates the application of Table I:

Sample designator: 14-33-24d

Interpretation:

14- designates aerial duplicating film

33- designates polyester base, not less than 3.9 or more than 5.1 mils thick

24- designates blue sensitive emulsion capable of attaining a gamma of at least 1.51 but not greater than 3.00, when processed as specified.

d- designates a sensitivity of not less than 4.3 nor more than 8.5 when determined as specified herein.

S5. TEST EQUIPMENT AND MATERIALS

S5.1 Film quality determination. All photographic black-and-white film shall be tested in accordance with the equipment, methods, and other provisions of this standard and the applicable detail specifications for determining the acceptable quality of the film.

S5.2 Apparatus

S5.2.1 Densitometer. The densitometer, used in these tests, shall have a repeatable accuracy of ± 0.02 in the density range from 0 to 1.00 and ± 0.03 in the density range from 1.00 to 3.00. The densitometer shall provide type Vl-b density as specified in ANSI PH2.19.

S5.2.2 Sensitometer. The sensitometer, used in these tests, shall be of the nonintermittent, intensity-scale type, with calibrated lamps and variable exposure capability. The sensitometer shall have a repeatable accuracy of ± 0.025 log E at the 11th step of the exposure modulator or the equivalent area of a continuous exposure modulator.

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TABLE I. FILM CLASSIFICATION SYSTEM

SAMPLE DESIGNATOR = 1 4 - 3 3 - 2 4 d

Product	Type	Support	Total Thickness (Mils)	Spectral Sensitivity	Gamma	Relative Sensitivity
1 Aerial film	1 Negative	1 Cellulose-ester regular	1 1.0 - 2.4	1 Extended ultra-violet	1 0.10 0.50	a .57 - 1.1
2 Motion picture film	2 Positive film	2 Cellulose-ester low shrink	2 2.5 - 3.8	2 Blue sensitive	2 0.51 1.00	b 1.2 - 2.1
3 Roll and film pack	3 Reversal	3 Polyester	3 3.9 - 5.1	3 Ortho-chromatic	3 1.01 1.50	c 2.2 - 4.2
4 Sheet film	4 Duplicating	4 Polystyrene	4 5.2 - 6.5	4 High Green	4 1.51 3.00	d 4.3 - 8.5
5 Photo-mechanical	5 Sound recording	5 Stripping base	5 6.6 - 8.2	5 Panchromatic	5 3.01 5.00	e 8.6 - 17
6 Micro-film	6 Other		6 8.3 - 10.0	6 Panchromatic (extended red)	6 5.01 above	f 18 - 35
			7 10.1 - 12.0	7 Infrared	7 Other	g 36 - 71
			8 Other			h 72 - 140
						i 141 - 280
						j 281 - 570
						k 571 - 1100
						l 1101 - 2300
						m 2301 - 4500

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S5.2.2.1 Exposure modulation. The sensitometric step tablet or continuous wedge shall be as described in the "modulation" paragraph of ANSI PH2.5. The total transmission density of the step tablet, or continuous wedge, plus any "neutral-density" filters shall not vary more than ± 0.08 throughout the wave length range of 350 nanometers (nm) to 900 nanometers.

S5.2.2.2 Illuminant No. 1. Illuminant No. 1 shall be as specified in the appropriate paragraphs of ANSI PH2.29.

S5.2.2.3 Illuminant No. 2. Illuminant No. 2 shall be a tungsten lamp operating at a color temperature of 2,850° Kelvin (K) to 3,000° K. An optimum illuminance shall be selected which will yield a characteristic curve from which the parameters, required by the detail specification, can be obtained.

S5.2.2.4 Illuminant No. 3 (Infrared (I.R.)). Illuminant No. 3 shall be the same as that specified in S5.2.2.3 for Illuminant No. 2, except that filters with a spectral range equivalent to the Kodak Wratten No. 25 (A) or the Corning Glass Co. No. 3966 (having a density of 0.3 at 700 nanometers), or equal, will be inserted in the light path. An optimum irradiance shall be selected, which will yield a characteristic curve from which the parameters, required by the detail specification, can be obtained.

S5.2.3 Dimensional change gage. The dimensional change gage shall be the "pin gage" or optical gage described in ANSI PH1.32 or a gage of at least equal accuracy.

S.5.2.4 Constant humidity chamber. A room or cabinet, which meets the requirements of section 4, ANSI PH1.32 shall be used for those tests requiring controlled conditions of temperature and relative humidity.

S5.2.5 Processing. When using the processing solutions, processing times and temperature specified in the detail specifications, the sensitometric results obtained and their reproducibility shall be equivalent to those obtained when using the method specified in ANSI PH2.5. Unless otherwise specified, use only freshly mixed developers.

S5.3 Test processing solutions.

S5.3.1 Formulas. The applicable formulas, as specified herein, shall apply in the mixing of processing solutions for testing of films covered by this standard. The grade of the chemicals used shall be as specified in O-C-275. The premeasured chemistry used shall be as specified in the detail specification.

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S5.3.1.1 Developer No. 1, high contrast. This formula shall have the following composition, corresponding to MIL-D-4824 (Armed Forces Developer No. 1).

Water (de-ionized) about 90°F	750.0 milliliters (ml)
p-Methylaminophenol sulfate	2.0 grams (g)
Sodium sulfite (anhydrous)	90.0 g
Hydroquinone	8.0 g
Sodium carbonate (monohydrate)	52.5 g
Potassium bromide	5.0 g
Water (de-ionized) to make	1.0 liter

S5.3.1.2 Developer No. 2, medium contrast. This formula shall have the following composition as given in ANSI PH2.5.

Water (de-ionized) about 125°F	500.0 ml
p-Methylaminophenol sulfate	1.0 g
Sodium sulfite (anhydrous)	25.0 g
Hydroquinone	2.0 g
Sodium carbonate (anhydrous)	3.0 g
Potassium bromide	0.38 g
Water (de-ionized) to make	1.0 liter

S5.3.1.3 Developer No. 3, low contrast. This formula shall have the following composition, corresponding to MIL-D-4826 (Armed Forces Developer No. 3):

Water (de-ionized) about 90°F	750.0 ml
p-Methylaminophenol sulfate	2.0 g
Hydroquinone	5.0 g
Sodium sulfite (anhydrous)	100.0 g
Sodium tetraborate (decahydrate)	2.0 g
Water (de-ionized) to make	1.0 liter

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S5.3.1.4 Motion picture negative developer. This developer shall have the following composition.

Water (de-ionized) about 90°F	750.0 ml
p-Methylaminophenol sulfate	1.5 g
Hydroquinone	1.3 g
Sodium sulfite (anhydrous)	75.0 g
Sodium tetraborate (decahydrate)	4.5 g
Potassium bromide	0.4 g
Water (de-ionized) to make	1 liter

S5.3.1.5 Positive print developer. This developer shall have the following composition:

Water (de-ionized) about 90°F	750.0 ml
p-Methylaminophenol sulfate	0.3 g
Sodium sulfite (anhydrous)	38.0 g
Hydroquinone	6.0 g
Sodium carbonate, monohydrated	22.5 g
Potassium bromide	0.9 g
Citric acid	0.7 g
Potassium metabisulfite	1.4 g
Water (de-ionized) to make	1.0 liter

S5.3.1.6 Photomechanical film developer. This developer shall not be used until 2 hours after mixing and should be discarded 24 hours after mixing. This developer shall have the following composition.

Water (de-ionized) about 90°F	600.0 ml
Sodium sulfite (anhydrous)	30.0 g
Paraformaldehyde	7.5 g
Sodium bisulfite (anhydrous)	2.2 g
Boric acid (crystals)	7.5 g
Hydroquinone	22.5 g
Potassium bromide	1.6 g
Water (de-ionized) to make	1.0 liter

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S5.3.1.7 Reversal processing chemistry. This chemistry shall have the following composition.

First developer:

Warm water (de-ionized) (about 125°F or 52°C)	750.0 ml
Sodium hexametaphosphate	0.5 g
p-Methylaminophenol sulfate	0.6 g
Sodium sulfite (anhydrous)	50.0 g
Hydroquinone	20.0 g
Potassium bromide	4.0 g
Potassium thiocyanate	5.0 g
Sodium hydroxide (pellets)	20.0 g
Water (de-ionized) to make	1.0 liter

Bleaching bath:

Water (de-ionized)	1.0 liter
Potassium dichromate	9.5 g
Sulfuric acid (concentrated)	12.0 ml

Clearing bath:

Sodium sulfite (anhydrous)	90.0 g
Water (de-ionized) to make	1.0 liter

Second developer:

Warm water (de-ionized) (about 125°F or 52°C)	750.0 ml
Sodium sulfite (anhydrous)	45.10 g
Hydroquinone	22.5 g
Boric acid, crystals	5.5 g
Potassium bromide	2.5 g
Sodium hydroxide (caustic soda)	22.5 g
Water (de-ionized) to make	1.0 liter

S5.3.1.8 Sodium based fixing bath. This fixing bath shall have the following composition corresponding to MIL-F-26615:

Water (de-ionized) about 90°F	600.0 ml
Sodium thiosulfate (crystalline)	240.0 g
Sodium sulfite (anhydrous)	15.0 g
Acetic acid (28 percent)	50.0 ml
Boric acid (crystals)	7.5 g
Aluminum potassium sulfate	15.0 g
Water (de-ionized) to make	1.0 liter

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S5.3.1.9 Ammonium based fixing bath. This fixing bath shall have the following composition corresponding to MIL-F-26615:

Water (de-ionized) about 90°F	500.0 ml
Ammonium thiosulfate (60 percent)	225.0 ml
Sodium sulfite (anhydrous)	15.0 g
Acetic acid (28 percent)	50.0 ml
Boric acid (crystals)	7.5 g
Aluminum potassium sulfate	15.0 g
Water (de-ionized) to make	1 liter

S6. TEST METHODS AND TESTS

S6.1 General description. The following is a general description of the methods which shall be used to determine the sensitometric and physical properties of the film to be tested. Only those tests which are necessary to determine the characteristics for which specific values or ranges are given in the applicable detail specification shall be performed.

S6.2 Determination of sensitometric characteristics. The sensitometric characteristics of aerial film, roll film, film pack, sheet film and plates intended for conventional daylight photography shall be determined by the methods and procedures specified in ANSI PH2.34 and ANSI PH2.5, respectively, except that sensitometric processing machines with nitrogen burst or reciprocal paddle agitation may be used, provided that they have been calibrated against the ANSI PH2.5 method. The sensitometric characteristics of other films shall be determined by the following steps, as applicable, or as specified in the detail specification.

S6.2.1 Test strips. Unless otherwise specified, test strips shall be 35 by 250 millimeters (mm) except for 16 mm film, in which case they shall be 16 by 250 mm.

S6.2.1.1 Conditioning of sensitometric strips. The test samples shall be in equilibrium with an atmosphere maintained at a temperature of 20° ± 5°C (68° ± 9°F) and a relative humidity of 50 ± 10 percent.

S6.2.1.2 Exposure of sensitometric strips. The exposure shall be nonintermittent. The time of exposure and illuminant shall be as stated in the detail specification. Two film strips shall be exposed on the sensitometer for each condition specified in the applicable detail specification.

S6.2.1.3 Development of sensitometric strips. Unless otherwise specified in detail specifications, the processing temperature shall be 20° ± 0.5°C (68° ± 0.9°F). The

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developing time shall be as specified in the applicable detail specification ± 1.0 percent. Sensitometric processing machines with nitorgen burst or reciprocating paddle agitation may be used provided they have been calibrated against the ANSI PH2.5 method. Unless otherwise specified, use only freshly mixed developers.

S6.2.1.4 Fixing. The test samples shall be immersed in a sodium or ammonium based fixing bath. The fixing bath shall be held at 20° ± 5°C (68° ± 9°F), and the fixing time shall be at least twice the clearing time but no longer than 15 minutes. The samples shall be agitated vigorously during the first 30 seconds in the fixing bath.

S6.2.1.5 Washing and drying of sensitometric strips. All test strips shall be washed and dried as described in ANSI PH2.5 and as specified in the detail specification.

S6.2.1.6 Preparation of "fixed out" strips. Two unexposed strips shall be fixed, washed, and dried as specified above. These strips shall then be used to determine base density. If the antihalation dye is not bleached or removed by the above process, the test strips shall be processed in a developer solution lacking the developing agent, then fixed, washed, and dried as specified above.

S6.2.1.7 "Fog" strips. Two unexposed test strips shall be developed, fixed, washed, and dried as specified herein. These strips shall be used to determine fog density.

S6.2.1.8 "Incubation fog" strips. Two test strips shall be conditioned for 3 hours to 50 ± 10 percent relative humidity and 68 ± 9°F. The conditioned strips shall then be placed in a photographically inert, lighttight, water vapor impervious container so that the emulsion does not contact the container. The container shall be of such a size that the ratio of the volume of air to the volume of the film will be less than 6 to 1. The container shall then be placed in an oven with the temperature maintained at 120° ± 5°F for 7 days, then removed and allowed to cool to room temperature. The strips shall be developed, fixed, washed, and dried as specified herein. These strips shall be used to determine incubation fog density.

S6.2.1.9 Filter factor. Two test strips shall be exposed on the sensitometer as specified herein, except the filters as specified in the detail specifications, shall be placed in the light path. If necessary, exposure times of a longer duration than those required for a normal exposure shall be used. The exposure time shall be indicated on the data sheet shown on figure 1. The exposed strip shall be developed, fixed, washed and dried as specified herein.

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S6.2.1.9.1 Determination of filter factor. The filter factor is the ratio of the relative sensitivity when exposed and processed under specified conditions to the relative sensitivity when exposed with the addition of the particular filter in the light path and processed under the same condition.

S6.2.1.10 Densitometry. The total diffuse transmission density of each step on each pair of test strips shall be measured. The average density of each step shall be recorded. The total diffuse transmission density for the "fixed out" strips, "fog" strips, and "incubation fog" strips shall be determined as the average density of each pair of strips determined by measuring the density of five areas on each strip.

S6.2.1.11 Plotting the characteristic curve. The average diffuse transmission density of each step on the film strip shall be plotted against the corresponding logarithm of the exposure on a test sheet similar to that shown on figure 1 The logarithm is to the base 10 and the exposure is expressed in lux-seconds or ergs per square centimeter. The actual densities plotted and all construction lines used to determine relative sensitivity, gamma, etc., shall be clearly shown.

S6.2.1.12 Determination of average gradient

S6.2.1.12.1 Photomechanical films. The average gradient of photomechanical films is the slope of the straight line drawn between points E_1 and E_2 on the characteristic curve. E_1 shall be a point on the curve where the density is 0.10 above base plus fog density. Point E_2 shall be a point on the curve where the log exposure is 0.40 greater than the log exposure at point E_1. If point E_2 exceeds 2.5 in density above base plus fog density, a point on the characterisitc curve at a density of 2.5 above base plus fog shall be used.

S6.2.1.12.2 Motion picture reversal films. The average gradient of motion picture reversal films is the slope of the straight line drawn between points E_1 and E_2 on the characteristic curve. E_1 shall be a point on the curve at a total density of 0.5 and point E_2 shall be a point on the curve at a total density of 2.0.

6.2.1.12.3 Microfilm. The average gradient of microfilm is the slope of the straight line drawn between points E_1 and E_2 on the characteristic curve. Points E_1 and E_2 shall be the points on the curve where the densities are 0.10 and 1.20 greater than the base plus fog density, respectively.

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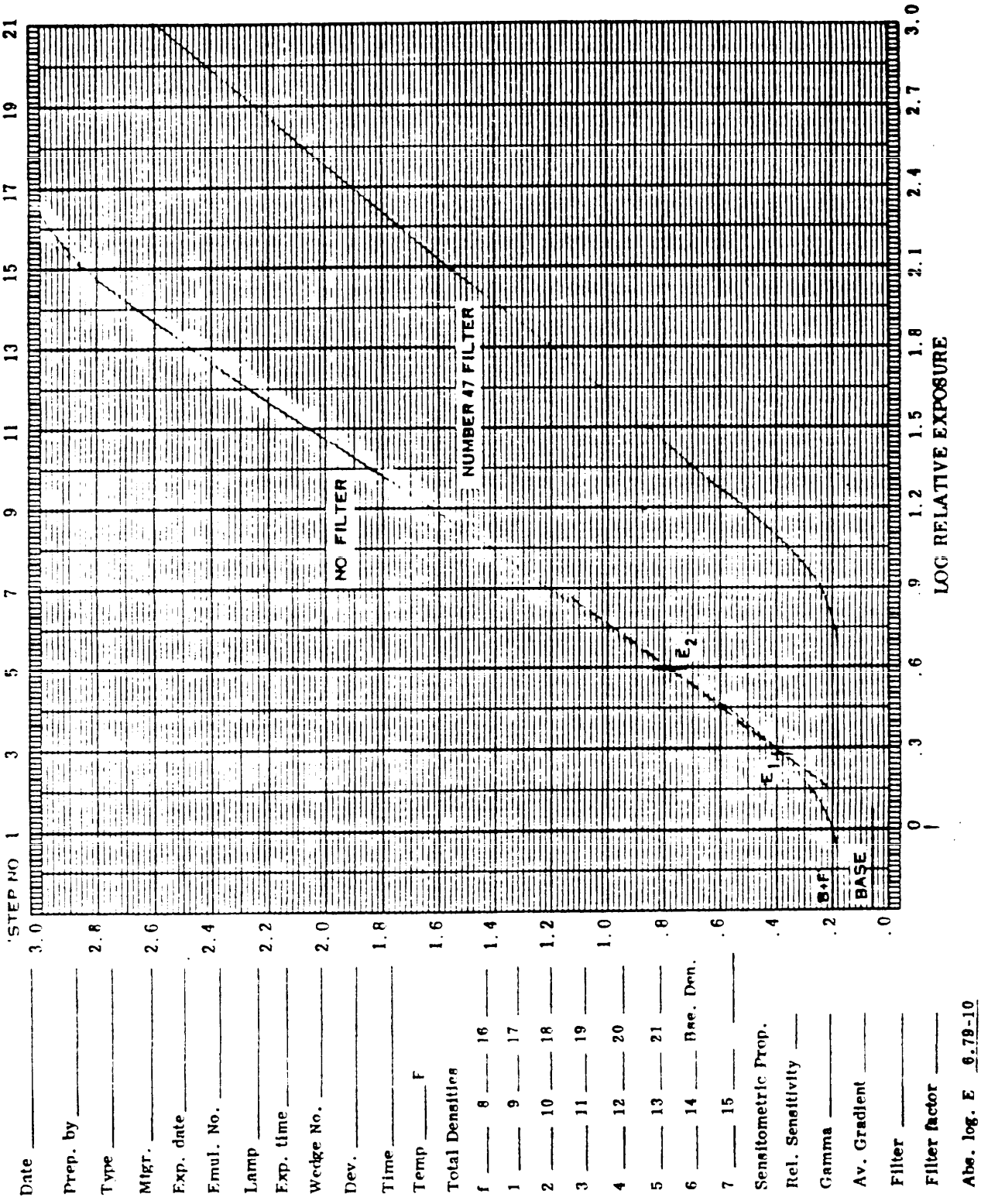


FIGURE 1

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S6.2.1.13 Tabulation of sensitometric data. All sensitometric data obtained from tests conducted to determine compliance with the sensitometric requirements, as stated in the applicable detail specification, shall be tabulated on the test sheet as indicated on figure 1. Any departure from the test procedure called for in this standard shall be noted.

S6.2.2 Resolving power test. This test shall be performed with the equipment and procedure in accordance with ANSI PH2.33 except that the processing shall be as specified in the detail specification.

S6.3 Dimensional change characteristics

S6.3.1 Humidity coefficient test. The procedures, calculations, and reporting of this test shall be in accordance with ANSI PH1.32. The humidity coefficient shall be measured on unprocessed film.

S6.3.2 Dimensional change due to processing test. Tests for dimensional change characteristics due to processing shall be conducted in accordance with procedures II and III of ANSI PH1.32. The test samples shall be processed to maximum density.

S6.3.3 Dimensional change due to processing plus aging tests. These tests shall be conducted on support types 1 and 2 (see Table I) and reported in accordance with ANSI PH1.32. The aging conditions shall be 120° ± 2°F and 20 ± 2 percent relative humidity. The aging time shall be one week.

S6.4 Brittleness test. Unless otherwise specified in the detail specification, the brittleness test shall be in accordance with method "B" of ANSI PH1.31. The testing temperature shall be 70° ± 1°F, and the testing relative humidity shall be 14 ± 1 percent.

S6.5 Curl test. The film shall be tested for curl in accordance with ANSI PH1.29.

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S6 .6 Tensile strength test. The film shall be subjected to the tensile strength test Method 1013, of Fed. Test Method Std. No. 406. The number of test samples and minimum permissible strength shall be as specified by the detail specification.

S6.7 Thermal coefficient of expansion test. The film shall be subjected to the thermal coefficient of expansion test as described in section 7 of ANSI PH1.32., using Fahrenheit temperature. The number of test samples shall be as specified in ANSI PH1.32 and maximum permissible coefficient (%/°F) shall be as specified by the detail specification.

S6.8 Physical defects test. Film samples, as specified by the detail specification, shall be exposed to a uniform light source to produce a density upon development of 0.7 to 1.0 and processed in an appropriate developer. The film shall be examined visually to determine compliance with the specification.

S7 NOTES

S7.1 International standardization. Certain provisions of this standard are the subject of international standardization agreement STANAG-3182 and STANAG-3188. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement(s) concerned, the preparing activity will take appropriate action through international standardization channels, including departmental standardization offices, if required.

S7.2 Changes. When a Federal agency considers that this standard does not provide for its essential needs, written request for changing or adding to the standard supported by adequate justification, shall be submitted to the Administration. This justification will explain wherein the standard does not provide for essential needs. The request should be sent in duplicate to the General Services Administration, Federal Supply Service, Standardization Division, Washington, D.C. 20407. The Administration will determine the appropriate action to be taken and will notify the agency.

S7.3 Conflict with referenced specification. Where the requirements specified in this standard conflict with any requirements

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in a referenced specification, the requirements of this standard shall apply. Nature of conflict between the standard and the referenced specification shall be submitted in duplicate to the General Services Administration, Federal Supply Service, Standardization Division, Washington, D.C. 20407.

Custodians:
Army - EL
Navy - AS
Air Force - 70

Review activities:
Army - EL, MI, AV
Navy - AS
Air Force - 11

User activity:
DGSC - GS

Preparing activity:
Air Force - 70

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